Interpretable Activation-Selection Neural Networks for Symbolic Regression of Parameter-Dependent Hamiltonian Eigenvalues

Alexander Yu. Kharin[1]*, Kirill F. Sheberstov[2]*

[1]*InfoMediji LTD*

[2] *Chimie Physique et Chimie du Vivant (CPCV, UMR 8228), Department of Chemistry, École Normale Supérieure, PSL University, Sorbonne University, CNRS, Paris, 75005, France*

Contact authors:

alexankhar@infomediji.com

kirill.sheberstov@ens.psl.eu

ORCID:

AYK: 0000-0002-0147-0384

KFS: 0000-0002-3520-6258

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Abstract

Analytical approximations to eigenvalues of parameter-dependent Hamiltonians can provide physical insight that is not readily apparent from numerical diagonalization alone. Here, we introduce an activation-selection network (ASN), a differentiable symbolic-regression architecture in which each input node learns a sparse combination of predefined analytical functions and the trained network can be converted directly into an explicit expression. Before regression, the Hamiltonian parameters and eigenvalues are expressed as dimensionless ratios. This normalization enforces dimensional homogeneity, reduces the number of independent variables, and ensures that the extracted expressions do not depend on the choice of energy units. Using the library $\{0, x, x^2\}$, compositions across successive hidden layers generate polynomial expansions of progressively higher degree; polynomial expansions of arbitrary finite degree can therefore be obtained in principle by increasing the network depth. We apply the ASN to effective three- and four-site spin-chain Hamiltonians relevant to zero-quantum nuclear magnetic resonance. Comparisons with degenerate perturbation theory show that the extracted expressions capture the expected constant, linear, and quadratic structure. These results establish the ASN as a framework for selecting compact symbolic representations when several functional forms are plausible, while showing that adaptive activation selection does not provide an intrinsic accuracy advantage over a suitable predefined basis.

Keywords: Symbolic Regression, Interpretable Neural Networks, Activation Selection, Hamiltonian Eigenvalues, Perturbation Theory, Nuclear Magnetic Resonance

I. Introduction

Feedforward neural networks are universal approximators of continuous functions [1] and are widely used to model physical systems [2]. Their predictive flexibility, however, does not by itself provide compact analytical expressions relating physical variables. By contrast, symbolic regression approach seeks explicit mathematical expressions that balance approximation accuracy and structural simplicity. In physics, such expressions can reveal scaling laws, symmetries, limiting behavior, and conservation laws that may remain obscured in purely numerical models. Earlier differentiable approaches to equation learning include Equation Learner (EQL) models, which combine predefined analytical operators with sparsity regularization to identify interpretable equations [3,4]. More recently, Kolmogorov-Arnold networks (KANs) introduced a different neural architecture based on learnable univariate functions [5].

Specifically, KANs replace the scalar weights of conventional multilayer perceptrons (MLPs) with trainable univariate functions placed on the connections between nodes. In the original KAN framework, these edge functions are represented by splines and can subsequently be matched to and replaced by functions from a predefined symbolic library [5]. The activation-selection network (ASN) introduced here follows a different construction. For each node serving as an input to a layer, training selects one function from a predefined set of analytical functions. Each ASN layer contains two sets of trainable weights: one controls activation selection, while the other

defines the linear transformation of the activated node outputs, as in a conventional MLP. The trained ASN can therefore be converted directly into compact symbolic expressions without requiring a separate symbolic approximation of learned spline functions. This construction is suited to physical problems for which numerical solutions can be calculated readily, whereas compact analytical approximations remain difficult to derive.

The effectiveness of an ASN therefore depends on selecting a function library consistent with the expected analytical structure of the target problem. For Hamiltonian eigenvalue problems, perturbation theory provides such guidance when the Hamiltonian can be separated into a dominant part and one or more smaller perturbations [6]. The eigenvalues can then be expanded in powers of the corresponding perturbation parameters, with truncation of these series yielding polynomial approximations. This motivates the polynomial function library used in the present work. The ASN framework is not, however, intrinsically restricted to polynomial functions.

Here, we apply the ASN to symbolic regression of parameter-dependent Hamiltonian eigenvalues. The network uses the activation set $\{0, x, x^2\}$, where 0 denotes the zero function, and compositions across successive layers can generate polynomials of increasing degree. As controlled benchmarks, we consider effective spin-chain Hamiltonians relevant to zero-quantum nuclear magnetic resonance (NMR) [7–9]. Their eigenvalues can be calculated numerically to provide exact reference data, whereas compact parameter-dependent analytical approximations become increasingly difficult to derive for longer or nonuniform chains. Expressing the Hamiltonians in terms of dimensionless ratios of original parameters reduces the number of independent variables and ensures dimensional homogeneity. For the uniform three-site chain, we compare the ASN expressions with results from degenerate perturbation theory. We then derive a two-parameter perturbative reference for the nonuniform three-site chain and extend the ASN analysis to a four-site chain. Fixed-basis least-squares polynomial fits provide an additional baseline for these comparisons. Together, these benchmarks test whether learned activation selection can yield compact analytical approximations beyond direct perturbative and least-squares approaches.

II. Activation-selection neural-network framework

The goal of symbolic regression is to represent a target relationship using a mathematical expression optimized jointly for predictive accuracy and simplicity. The ASN implements this objective by combining a differentiable choice among predefined analytical activation functions with linear transformations between successive layers (**Fig. 1**).

The structure of a single ASN layer is illustrated in **Fig. 1(a)**. Each layer comprises an activation-selection sublayer followed by a linear-transformation sublayer. The activation-selection sublayer maps each input x_i to a transformed variable v_i by forming a differentiable combination of k predefined activation functions. The subsequent linear-transformation sublayer combines these variables to produce the outputs y_l. The activation-selection parameters and linear weights are optimized jointly.

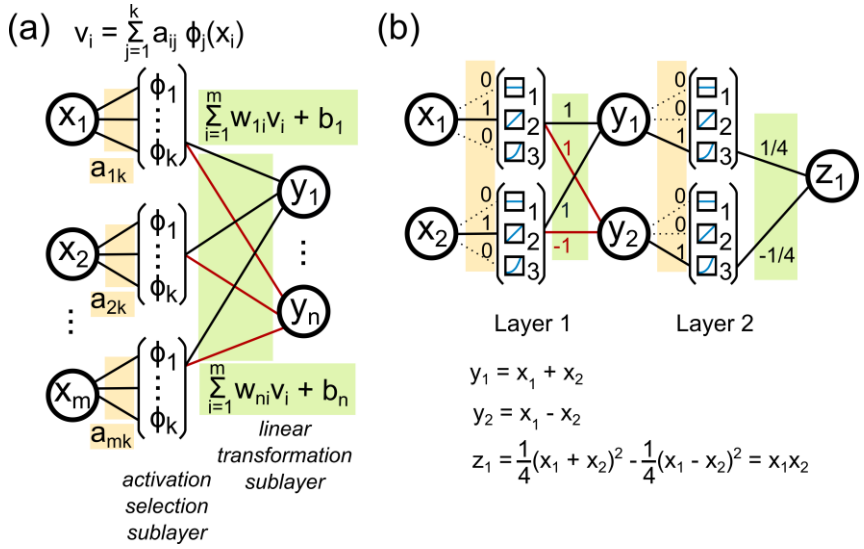


Fig. 1. Architecture of the activation-selection network (ASN). (a) Structure of a single ASN layer. For each input x_i, k candidate activation functions $\phi_j(x_i)$ are evaluated and combined using the softmax-normalized coefficients a_{ij}. An $L_{0.5}$-type penalty favours nearly one-hot coefficient vectors, thereby promoting the selection of a single activation for each input node. The transformed variables v_i are then combined by a linear transformation with weights w_{li} and biases b_l to produce the output nodes y_l. (b) Two-layer ASN that exactly implements multiplication of two inputs. The first layer forms $y_1 = x_1 + x_2$ and $y_2 = x_1 - x_2$. The second layer selects the quadratic activation and combines the resulting terms as $z_1 = \frac{1}{4}(x_1 + x_2)^2 - \frac{1}{4}(x_1 - x_2)^2 = x_1x_2$. The candidate set $\{0, x, x^2\}$, where 0 denotes the zero function, permits higher-degree polynomials to be constructed by stacking layers and allows selected inputs to be suppressed.

To define the activation-selection step mathematically, each input node x_i is assigned a vector of trainable logits $\boldsymbol{\alpha}_i = (\alpha_{i1}, \ldots, \alpha_{ik})$, with one logit for each candidate activation function. Training proceeds in two stages. During the initial stage, the logits are converted into activation-selection coefficients using the softmax transformation,

$$a_{ij} = \mathrm{softmax}(\boldsymbol{\alpha}_i)_j = \exp(\alpha_{ij}) \,/\left[\sum_{r=1}^{k} \exp(\alpha_{ir})\right]. \tag{1}$$

The activation-selection coefficients are nonnegative and normalized in both stages:

$$a_{ij} \geq 0, \sum_{j=1}^{k} a_{ij} = 1. \tag{2}$$

The softmax parameterization allows all candidate activation functions to contribute through differentiable coefficients, enabling the logits to be optimized jointly with the linear weights. To favor nearly one-hot coefficient vectors, an $L_{0.5}$-type regularization term is minimized for each input node during the softmax stage:

$$L_{0.5}^{(i)} = \sum_{j=1}^{k} \sqrt{a_{ij}} - 1. \tag{3}$$

The total regularization term is obtained by summing $L_{0.5}^{(i)}$ over all input nodes in all ASN layers. Because the coefficients are nonnegative and sum to unity, the sum of their square roots

approaches its minimum when one coefficient approaches unity and the remaining coefficients approach zero. Subtracting 1 sets this limiting minimum to zero without changing its location or the gradients of the regularization term.

During the final stage of training, softmax is replaced by sparsemax [10]:

$$a_{ij} = \mathrm{sparsemax}(\boldsymbol{\alpha}_i)_j = \max(\alpha_{ij} - \tau(\alpha_i), 0). \quad (4)$$

Here, $\tau(\alpha_i)$ is a threshold calculated independently for each input i such that the coefficients satisfy Eq. (2). Unlike softmax, sparsemax can assign coefficients of exactly zero to unused activation functions, thereby producing a sparse analytical representation. The $L_{0.5}$-type regularization favors coefficient vectors in which one coefficient equals 1 and all others equal 0. A final trained sublayer therefore corresponds to the selection of a single library function, such as 0, x, or x^2.

The effective activation corresponding to input x_i is given by:

$$v_i = \sum_{j=1}^{k} a_{ij}\phi_j(x_i). \quad (5)$$

Finally, the linear-transformation sublayer combines these activations using trainable weights and biases, as in a conventional feedforward layer. For an ASN layer with m input nodes and n output nodes, the output y_ℓ is given by:

$$y_\ell = \sum_{i=1}^{m} w_{\ell i} v_i + b_\ell \,, \ \ell \in \{1, \ldots, n\}. \quad (6)$$

Fig. 1(b) demonstrates that the ASN can represent the product of two arbitrary input variables exactly using only the activation set $\{0, x, x^2\}$. Because the output of one ASN layer serves as the input to the next, repeated application of the layers can generate products and progressively higher powers. For an ASN with L layers, the polynomial degree can at most double at each layer, giving an upper bound of 2^L. A two-layer ASN can therefore represent quartic terms in principle. In the specific construction shown in **Fig. 1(b)**, however, the first layer uses identity activations to form linear combinations, and only the second layer applies the quadratic activation. Explicit logits, weights, and biases for this construction are given in Appendix A. The construction in **Fig. 1(b)** is not unique: wider networks can represent the same product, while unnecessary paths can be suppressed by selecting the zero activation.

III. Spin-chain Hamiltonians and benchmark eigenvalue problems

A. Modelling nuclear spins in molecular chains

To evaluate whether the ASN can recover useful analytical approximations to Hamiltonian eigenvalues directly from numerical data, we consider effective spin-chain Hamiltonians

describing coupled methylene proton pairs in zero-quantum NMR experiments [7–9]. In these systems, locally prepared nuclear spin order can propagate along the molecular chain, providing an experimental realization of effective spin-chain dynamics [9]. The relevant eigenvalue differences determine the frequencies of the observed zero-quantum transitions, directly linking the regression targets to experimentally measurable quantities. Each methylene proton pair is represented by an effective spin-1/2 particle, with basis states corresponding to the central triplet and singlet states:

$$|\uparrow\rangle \equiv |\mathrm{T}_0\rangle = \frac{1}{\sqrt{2}}(|\alpha\beta\rangle + |\beta\alpha\rangle), \qquad (7)$$
$$|\downarrow\rangle \equiv |\mathrm{S}_0\rangle = \frac{1}{\sqrt{2}}(|\alpha\beta\rangle - |\beta\alpha\rangle),$$

Here, α and β denote the single spin-1/2 particle Zeeman states. A chain of N methylene groups is thereby mapped onto N coupled effective spins.

Under this mapping, the scalar-coupling Hamiltonian projected onto the $\{T_0, S_0\}$ manifold becomes an effective one-dimensional chain of interacting spin-1/2 particles [9]:

$$\hat{H} = J_{intra} \sum_{i=1}^{N}(\hat{I}_{i,z} - \frac{1}{4}\hat{1}) + \sum_{i=1}^{N-1}\frac{\Delta J_i}{2}(\hat{I}_i^+\hat{I}_{i+1}^- + \hat{I}_i^-\hat{I}_{i+1}^+) + \sum_{i=1}^{N-1}\frac{\Delta J_i}{2}(\hat{I}_i^+\hat{I}_{i+1}^+ + \hat{I}_i^-\hat{I}_{i+1}^-). \qquad (8)$$

Here, N is the number of sites in the chain, and J_{intra} is the intrapair coupling between the two protons of a CH_2 group, assumed to be identical for all sites. The parameter ΔJ_i denotes the difference between the inequivalent interpair scalar couplings connecting sites i and $i + 1$, as defined in Appendix B. The operator $\hat{I}_{i,z}$ is the z component of the effective spin operator at site i, $\hat{1}$ is the identity operator, and $\hat{I}_i^+$ and $\hat{I}_i^-$ are the corresponding raising and lowering operators, respectively. The first term in Eq. (8) describes the local energy splitting between the T_0 and S_0 states. The second describes exchange between neighbouring effective spins, whereas the third simultaneously flips both spins.

The construction of the reduced basis and the derivation of Eq. (8), including the relationship between ΔJ_i and the underlying proton–proton couplings, are detailed in Appendix B.

B. Three-site and four-site benchmark Hamiltonians

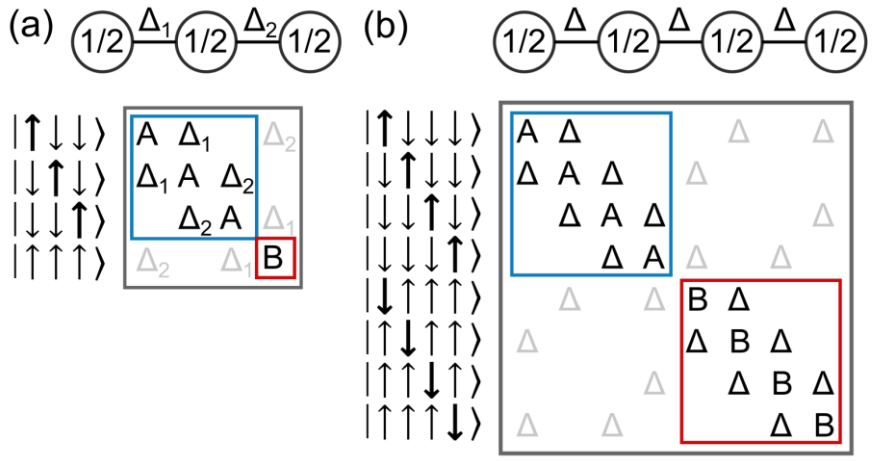


Fig. 2. Block structure of the Hamiltonian matrices for aliphatic spin chains containing three (a) and four (b) sites. Blue and red outlines indicate the manifolds with diagonal energies A and B, respectively. Black off-diagonal elements describe exchange between neighbouring sites, whereas grey elements connect the two manifolds through simultaneous neighbouring-site spin flips. For (a) $\Delta_1 = \Delta J_1/2$, $\Delta_2 = \Delta J_2/2$, A = -5/4 J_{intra}, B = 3/4 J_{intra}. For (b), $\Delta = \Delta J/2$, A = -2 J_{intra}, B = 0.

To construct benchmark eigenvalue problems of increasing complexity, we consider spin chains containing three and four sites. The Hamiltonian blocks whose eigenvalue differences determine the frequencies of observable transitions in the corresponding NMR spectra are shown in **Fig. 2** [9]. For the three-site chain, the two nearest-neighbour couplings ΔJ_1 and ΔJ_2 are treated independently, allowing both the uniform case, $\Delta J_1 = \Delta J_2$, and the nonuniform case, $\Delta J_1 \neq \Delta J_2$, to be examined. For the four-site chain, we consider uniform nearest-neighbour couplings, $\Delta J_1 = \Delta J_2 = \Delta J_3 = \Delta J$. Together, the uniform and nonuniform three-site cases and the uniform four-site case form a controlled progression from an analytically tractable reference problem to more general parameter-dependent Hamiltonians of increasing dimension.

C. Perturbative reference for the uniform three-site chain

For the uniform three-site chain, defined by $\Delta J_1 = \Delta J_2 = \Delta J$, the four eigenenergies were previously calculated to second order in ΔJ using time-independent perturbation theory for degenerate states [6,8]. The coupling hierarchy defining the perturbative regime is summarized in Appendix B. The resulting expressions provide an independent analytical reference for evaluating the ASN approximations:

$$E_1 = \frac{3}{4}J_{intra} + \frac{1}{4}\frac{\Delta J^2}{J_{intra}}, \quad (9a)$$

$$E_2 = -\frac{5}{4}J_{intra} + \frac{1}{\sqrt{2}}\Delta J - \frac{1}{8}\frac{\Delta J^2}{J_{intra}}, \quad (9b)$$

$$E_3 = -\frac{5}{4}J_{intra}, \quad (9c)$$

$$E_4 = -\frac{5}{4}J_{intra} - \frac{1}{\sqrt{2}}\Delta J - \frac{1}{8}\frac{\Delta J^2}{J_{intra}}. \quad (9d)$$

These expressions contain constant, linear, and quadratic terms in ΔJ and therefore provide a local analytical reference within the perturbative regime. For the nonuniform three-site chain, a corresponding two-parameter perturbative reference can be derived from the same degenerate manifold, as shown in Sec. III.D. The four-site problem then tests whether the ASN can obtain reproducible symbolic approximations for multiple eigenenergy branches of a larger Hamiltonian.

D. Nonuniform three-site chain: two-parameter perturbative structure

For the nonuniform chain, the unperturbed energy of the |↑↑↑⟩state is $3J_{\mathrm{intra}}/4$. The three states with one effective spin up and two down have the common unperturbed energy $-5J_{\mathrm{intra}}/4$. Within this three-state manifold, the first-order perturbation is represented by the exchange matrix

$$\frac{1}{2}\begin{pmatrix} 0 & \Delta J_1 & 0 \\ \Delta J_1 & 0 & \Delta J_2 \\ 0 & \Delta J_2 & 0 \end{pmatrix}.$$

Defining $S = \Delta {J_1}^2 + \Delta {J_2}^2$, we evaluate the second-order corrections due to coupling to |↑↑↑⟩ state in the eigenbasis of this exchange matrix:

$$E_1 = \frac{3}{4}J_{intra} + \frac{1}{8}\frac{S}{J_{intra}} + O\left(\frac{S^{3/2}}{J_{intra}^2}\right), \quad (10a)$$

$$E_2 = -\frac{5}{4}J_{intra} + \frac{\mathrm{sgn}(J_{intra})}{2}\sqrt{S} - \frac{1}{4}\frac{\Delta J_1^2 \Delta J_2^2}{J_{intra}S} + O\left(\frac{S^{3/2}}{J_{intra}^2}\right), \quad (10b)$$

$$E_3 = -\frac{5}{4}J_{intra} - \frac{1}{8}\frac{\left(\Delta J_2^2 - \Delta J_1^2\right)^2}{J_{intra}S} + O\left(\frac{S^{3/2}}{J_{intra}^2}\right), \quad (10c)$$

$$E_4 = -\frac{5}{4}J_{intra} - \frac{\mathrm{sgn}(J_{intra})}{2}\sqrt{S} - \frac{1}{4}\frac{\Delta J_1^2 \Delta J_2^2}{J_{intra}S} + O\left(\frac{S^{3/2}}{J_{intra}^2}\right). \quad (10d)$$

Here, $\mathrm{sgn}(J_{\mathrm{intra}}) = \frac{J_{\mathrm{intra}}}{|J_{\mathrm{intra}}|}$. The derivation is given in Appendix D. At $S = 0$, the rational terms containing $1/S$ are defined by continuity and assigned their zero-coupling limit of zero. Along the uniform line $\Delta J_1 = \Delta J_2 = \Delta J$, with $\Delta J/J_{\mathrm{intra}} \geq 0$, Eqs. (10a) - (10d) reduce to Eqs. (9a) - (9d) through second order. On a 101 × 101 grid covering $0 \leq \Delta J_1/J_{\mathrm{intra}}, \Delta J_2/J_{\mathrm{intra}} \leq 0.2$, the corresponding dimensionless eigenenergies reproduce exact diagonalization with a root-mean-square error (RMSE) of 4.28×10^{-5} and a maximum absolute error of 2.64×10^{-4}.

IV. Preparation of data for symbolic regression

A. Dimensionless formulation of the eigenvalue problem

Direct symbolic regression of the eigenenergies $E_i(J_{intra}, \Delta J_1, \Delta J_2)$ proved inefficient because the learned expressions generally violated dimensional consistency. Since both the Hamiltonian parameters and the eigenenergies are measured in units of frequency, physically meaningful analytical expressions must preserve the corresponding dimensionality. Because the Hamiltonian in Eq. (8) is linear in J_{intra} and ΔJ_i, multiplication of all coupling parameters by a common factor produces the same scaling of every eigenenergy. Because J_{intra} sets the overall energy scale, the three-site eigenvalue problem can be expressed in terms of two independent dimensionless coupling ratios, which we define as follows:

$$r_i \equiv \frac{\Delta J_i}{J_{intra}},\ i = 1,2;\ \varepsilon_k \equiv \frac{E_k}{J_{intra}},\ k = 1, \ldots, 4. \tag{11}$$

This normalization preserves dimensional consistency and transforms the original mapping $E_k(J_{intra}, \Delta J_1, \Delta J_2)$ into a two-dimensional regression problem $\varepsilon_k(r_1, r_2)$.

B. Assignment and symmetry of the dimensionless eigenenergy surfaces

A second challenge concerns the consistent assignment of the eigenenergy surfaces. For each pair (r_1, r_2), the Hamiltonian in Eq. (8) was diagonalized numerically. The resulting surfaces were labelled according to the convention $\varepsilon_1 > \varepsilon_2 \geq \varepsilon_3 \geq \varepsilon_4$, consistent with **Fig. 3** and Eqs. (9) after division by J_{intra}. Because physically $J_{intra} < 0$ [8,11,12], this ordering is the reverse of that of the signed physical eigenenergies, which satisfy $E_1 < E_2 \leq E_3 \leq E_4$. The adopted convention defines four single-valued dimensionless eigenenergy surfaces over the two-parameter space.

Along the uniform-chain line, $r_1 = r_2 = r$, Eqs. (9) describe branches that cross smoothly at $r = 0$, where E_2 and E_4 exchange their energy ordering. We initially used this branch assignment to construct the ASN training targets, but training was unsuccessful. Extending this assignment to the full two-dimensional coupling plane introduced discontinuities in the target energy surfaces: away from the degeneracy, the branches are separated as shown in **Fig. 3(c)**, so switching their labels produces finite jumps in the assigned energies. We therefore constructed the training targets by sorting the numerical eigenvalues into a fixed order at each parameter point. This yields continuous energy surfaces throughout the parameter domain. Along the uniform-chain line, sorting replaces the leading linear dependence on r of the two exchanging branches with a dependence on |r|, producing cusps at the degeneracy without introducing discontinuities as shown in **Fig. 3(b)**.

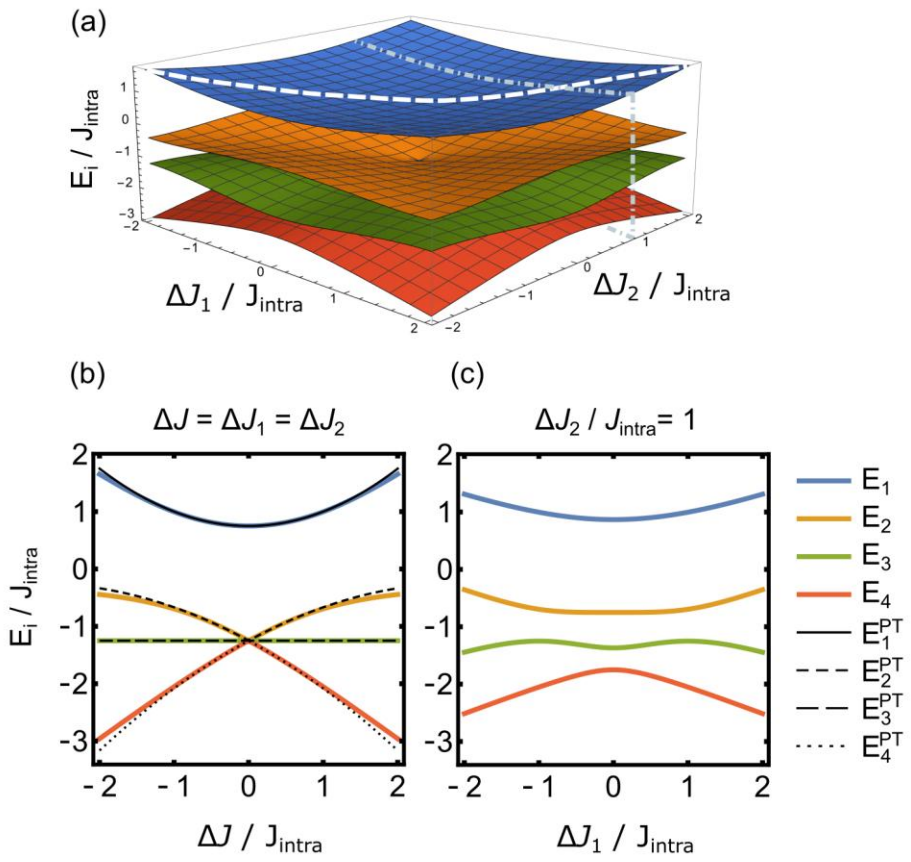


Fig. 3. Dimensionless eigenenergy surfaces and representative cross-sections for the Hamiltonian of the three-site spin chain. All energies and coupling constants are scaled by J_{intra}. (a) Numerically calculated eigenenergy surfaces E_i/J_{intra} as functions of the dimensionless couplings $\Delta J_1/J_{intra}$ and $\Delta J_2/J_{intra}$. The white dashed and the grey dot-dash lines indicate cross-sections used for (b) and (c), respectively. (b) Cross-section corresponding to the uniform chain, $\Delta J_1 = \Delta J_2 = \Delta J$. Coloured curves show the numerically calculated, energy-ordered eigenenergies, while the black solid, dashed, dash-dotted, and dotted curves show the second-order perturbative expressions in Eqs. (9). Across $\Delta J = 0$, the perturbative branches corresponding to E_2 and E_4 exchange their energy ordering. Maintaining a fixed ordering therefore introduces non-differentiability at the degeneracy point. (c) Cross-section for fixed $\Delta J_2/J_{intra} = 1$, illustrating the smooth variation of the eigenenergies away from the degeneracy. In the full two-dimensional parameter space, the eigenenergy surfaces remain ordered and should therefore be labelled according to their energies, $\varepsilon_1 > \varepsilon_2 \geq \varepsilon_3 \geq \varepsilon_4$, rather than by continuing the one-dimensional perturbative branches through $\Delta J = 0$.

The ordered eigenenergy surfaces exhibit symmetry properties that can be exploited to reduce the training domain. In particular, they are invariant under independent reversal of the sign of either coupling difference and under interchange of ΔJ_1 and ΔJ_2. The sign-reversal symmetry allows the ASN to be trained within the first quadrant, defined by $0 \leq r_1 \leq 2$ and $0 \leq r_2 \leq 2$, with the remaining quadrants subsequently reconstructed by reflection. The upper boundary of 2 was chosen to include sufficiently pronounced nonlinear behaviour while retaining a parameter range over which, along the uniform-chain line, the second-order perturbative expressions remain relatively close to the numerically calculated eigenenergies (**Fig. 3(b)**). This restriction places the non-differentiable point at the corner of the training domain and avoids requiring a polynomial network to reproduce the cusp across positive and negative coupling values. Because such non-differentiable behaviour cannot be represented exactly by a finite polynomial expansion, it would

otherwise present an additional challenge for symbolic regression based only on polynomial activation functions.

Restricting training to the first quadrant removes the sign-change cusp across positive and negative couplings, but it does not make the ordered spectrum analytic at the origin. Equations (10) show explicitly that the leading splitting contains $\sqrt{\Delta J_1^2 + \Delta J_2^2}$. A finite polynomial can therefore approximate but cannot exactly reproduce the local two-dimensional spectral geometry at the degeneracy.

V. Symbolic regression of spin-chain eigenenergies

A. Three-site spin chain

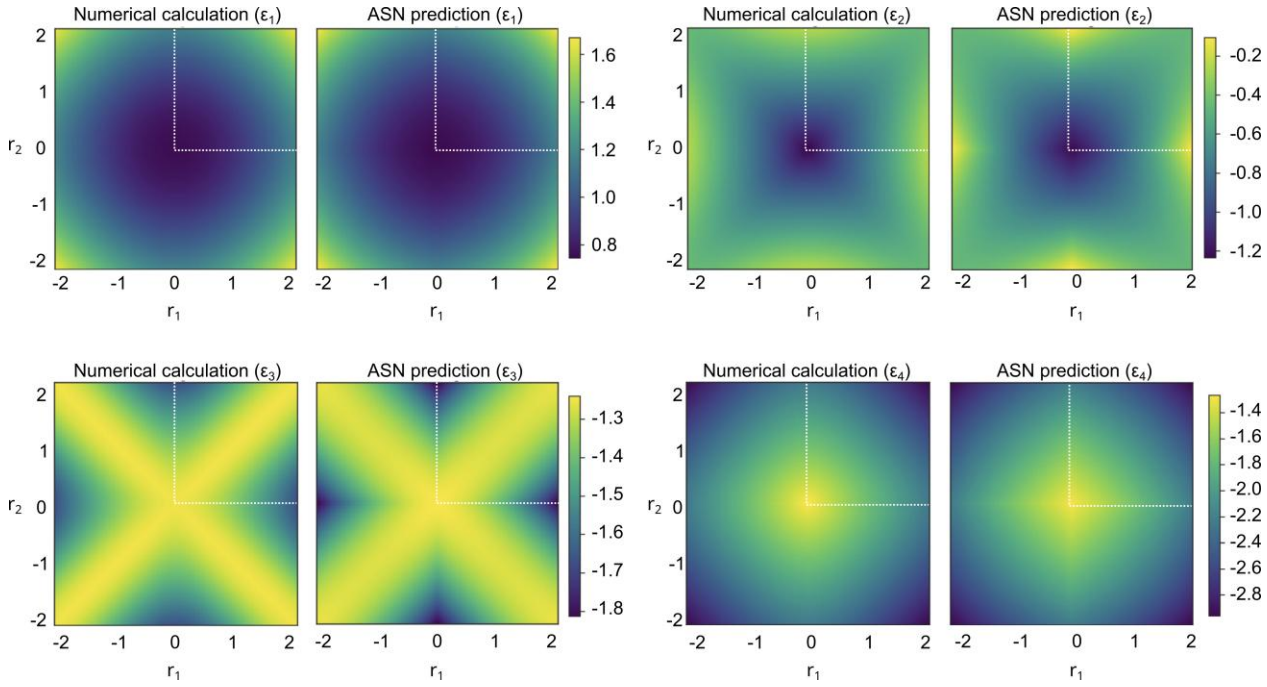


Fig. 4. Comparison of numerically calculated and ASN-predicted dimensionless eigenenergy surfaces for the three-site spin chain. Here, $r_i = \Delta J_i / J_{intra}$ and $\varepsilon_k = E_k / J_{intra}$. For each ε_k, the left and right panels show the numerical eigenenergies and corresponding ASN predictions, respectively. The ASN was trained within the first-quadrant domain $0 \le r_1, r_2 \le 2$, indicated by the white dotted lines, and the remaining quadrants were reconstructed using the independent sign-reversal symmetries of r_1 and r_2. A common colour scale is used for the numerical and predicted surfaces within each pair. The activation set used for training was $\{0, x, x^2\}$.

Having defined the dimensionless coupling ratios r_1 and r_2 and the ordered eigenenergy surfaces ε_i, we applied the ASN to the simultaneous symbolic regression of the four eigenenergy branches of the three-site chain. Candidate architectures with different depths and widths were compared in terms of predictive error and symbolic complexity, quantified by the total number of operations in the extracted expressions. The architecture (2, 16, 4), comprising two input nodes, one hidden

layer with 16 nodes, and four output nodes, was retained as a compact model balancing predictive accuracy and expression complexity.

The selected ASN was trained to predict the four dimensionless eigenenergies ε_i simultaneously using the activation set $\{0, x, x^2\}$. Inclusion of $|x|$ among the candidate activations was also examined but did not improve the prediction accuracy and was therefore omitted from the final activation set. The complete training procedure and hyperparameters are described in Methods.

Fig. 4 compares the numerically calculated dimensionless eigenenergy surfaces with the ASN predictions. Within the first-quadrant training domain, the ASN reproduces the curvature and overall topology of all four surfaces. Predictions in the remaining quadrants were obtained by reflection rather than by evaluating the polynomial network outside its training domain. Consequently, the cusp-like and cross-shaped structures in the full-domain representation arise from combining the first-quadrant approximation with the exact sign-reversal symmetries of the Hamiltonian. They should not be interpreted as nonanalytic behavior learned directly by the polynomial network.

For the selected ASN, the original training run yielded an overall RMSE of 1.79×10^{-2} across the complete sampled data set and all four outputs. The coefficients of determination were R^2 = 0.9994, 0.9888, 0.9483, and 0.9979 for ε_1–ε_4, respectively. These metrics were calculated from the numerical network outputs before coefficient thresholding, symmetry averaging, and rounding.

To obtain an interpretable representation, the trained ASN was converted into symbolic form, and the resulting expressions for ε_i were expanded in the dimensionless inputs r_1 and r_2. Multiplication by the signed J_{intra} then restored physical dimensions, $E_i^{ASN} = J_{intra}\varepsilon_i^{ASN}$, with the branch labels inherited from the dimensionless-energy ordering defined above. Because the interchange symmetry $r_1 \leftrightarrow r_2$ was not imposed explicitly during training, symmetry-equivalent coefficients were averaged after extraction and rounded to three decimal places. The expressions below therefore result from direct algebraic extraction followed by explicit post-processing steps: symmetry averaging and coefficient rounding.

$$E_1^{ASN} = 0.743J_{intra} + 0.022(\Delta J_1 + \Delta J_2) + 0.094\frac{(\Delta J_1)^2+(\Delta J_2)^2}{J_{intra}} + 0.023\frac{\Delta J_1 \Delta J_2}{J_{intra}}, \quad (12a)$$

$$E_2^{ASN} = -1.233J_{intra} + 0.361(\Delta J_1 + \Delta J_2) + 0.104\frac{(\Delta J_1)^2+(\Delta J_2)^2}{J_{intra}} - 0.372\frac{\Delta J_1 \Delta J_2}{J_{intra}}, \quad (12b)$$

$$E_3^{ASN} = -1.239J_{intra} - 0.028(\Delta J_1 + \Delta J_2) - 0.133\frac{(\Delta J_1)^2+(\Delta J_2)^2}{J_{intra}} + 0.288\frac{\Delta J_1 \Delta J_2}{J_{intra}}, \quad (12c)$$

$$E_4^{ASN} = -1.272J_{intra} - 0.355(\Delta J_1 + \Delta J_2) - 0.062\frac{(\Delta J_1)^2+(\Delta J_2)^2}{J_{intra}} + 0.054\frac{\Delta J_1 \Delta J_2}{J_{intra}}. \quad (12d)$$

Equations (12) provide compact global quadratic approximations to the dependence of the four eigenenergy branches on r_1 and r_2 within the first-quadrant training domain, $0 \le r_1, r_2 \le 2$. Unlike the local uniform-chain expressions in Eqs. (9a)–(9d), Eqs. (12a)–(12d) apply to nonuniform chains over a broad finite domain. Unlike the two-parameter perturbative expressions in Eqs. (10a)–(10d), however, their polynomial form does not reproduce the exact radial nonanalyticity at the degeneracy.

Along the uniform-chain line ($r_1 = r_2 = r$), the ASN expressions can be compared directly with the independently derived perturbative results (Eqs. (9)). The ASN reproduces their qualitative structure, including the nearly constant ε_3 branch and approximately opposite linear coefficients for ε_2 and ε_4. Specifically, the ASN coefficients of +0.722 and -0.710, respectively, are close to the perturbative values $+1/\sqrt{2}$ and $-1/\sqrt{2}$. The residual differences between the two sets of coefficients are expected because perturbation theory provides a local expansion about $r = 0$, whereas the ASN was trained to approximate all four eigenenergy surfaces over the full two-dimensional domain (r_1, r_2). The extracted ASN coefficients therefore describe global symbolic approximations and should not be interpreted as perturbative coefficients.

Because the ASN was trained directly on numerically calculated eigenenergies, its accuracy within the sampled domain is not constrained by the small-parameter requirement of perturbation theory. Along the uniform-chain line $0 \leq r \leq 2$, the rounded expressions in Eqs. (12a)–(12d) have an RMSE of 1.4 × 10^{-2} over the four eigenenergy branches, compared with 5.0 × 10^{-2} for the second-order perturbative expressions. Perturbation theory nevertheless remains substantially more accurate close to r = 0 and reproduces the constant ε_3 branch exactly. On an independent 121 × 121 grid covering the complete first-quadrant domain, the rounded ASN expressions yield an RMSE of 1.90 × 10^{-2}, compared with an RMSE of 1.79 × 10^{-2} for the unrounded network output. This small increase quantifies the loss of accuracy introduced by symbolic post-processing.

These results show that the ASN can extract compact global analytical approximations from numerical eigenenergy data. Predictive accuracy alone, however, does not establish an advantage of learned activation selection over direct polynomial regression. We therefore next compare the ASN expressions with fixed-basis least-squares polynomial fits.

B. Fixed-basis and physics-informed baselines

To determine whether the ASN provides an advantage over direct polynomial regression for this benchmark, we fitted ordinary least-squares models to the same 100,000 dimensionless input pairs. The symmetry-constrained quadratic basis $\{1,\ r_1 + r_2,\ r_1^2 + r_2^2,\ r_1 r_2\}$ gives a complete-sample RMSE of 1.62 × 10^{-2}, slightly below the value of 1.79 × 10^{-2} obtained for the unrounded ASN output. The fitted coefficients closely match those extracted from the ASN in Eqs. (12a)–(12d), showing that the network learns a global quadratic approximation that can also be obtained by direct linear regression once the appropriate basis has been specified.

Adding the physics-informed radial feature $\rho = \sqrt{r_1^2 + r_2^2}$ reduces the held-out RMSE to 1.40 × 10^{-2}. The improvement is particularly pronounced near the degeneracy: over $0 \leq r_1, r_2 \leq 0.2$, the radial-feature model gives an RMSE of 6.58 × 10^{-3}, compared with 1.74 × 10^{-2} for the symmetry-constrained quadratic model. The perturbative expressions in Eqs. (10a)–(10d) remain the most accurate in this local region, with an RMSE of 4.28 × 10^{-5}.

These comparisons clarify the role of the ASN in this benchmark. The network produces compact symbolic approximations, but it does not outperform ordinary least-squares regression when the

appropriate polynomial basis is known in advance. Its potential advantage lies instead in making the selection and composition of analytical building blocks part of the training procedure when several functional forms are plausible and the appropriate basis is not known beforehand.

C. Four-site spin chain

To extend the analysis beyond the three-site benchmark, we applied the ASN to the four-site chain shown in **Fig. 2(b)**, with all nearest-neighbor coupling differences set equal to ΔJ. Using J_{intra} as the energy scale, the eight dimensionless eigenenergies $\varepsilon_k = E_k/J_{intra}$ depend on the single dimensionless input $r = \Delta J/J_{intra}$. At $r = 0$, the eigenenergies form two fourfold-degenerate manifolds at $\varepsilon = 0$ and $\varepsilon = -2$. For nonzero r, the exchange terms split the levels within each manifold, while the double-quantum terms couple the two manifolds, producing a more complex eigenenergy structure than in the three-site benchmark [9]. An ASN with architecture (1, 10, 8), comprising one input, a single hidden layer with 10 nodes, and eight outputs, was trained to predict all eight eigenenergy branches simultaneously using the same zero, identity, and quadratic activation functions as in the three-site analysis.

Fig. 5 compares the numerically calculated eigenenergies with the ASN predictions. The network reproduces all eight branches over the sampled range $0 \le r \le 3$. The held-out validation RMSE was 7.9×10^{-3}. When evaluated over the complete sampled dataset, the per-branch coefficients of determination ranged from $R^2 = 0.9918$ to 0.9998. The largest deviations occur near the upper boundary of the sampled range, where the exact eigenenergies depart most strongly from quadratic behaviour.

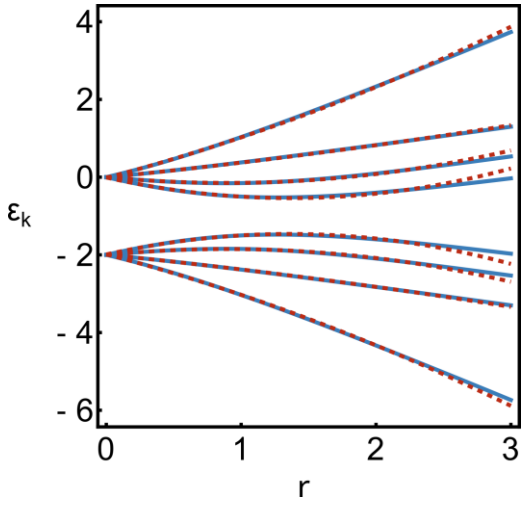


Fig. 5. Numerical and ASN-predicted eigenenergy branches for the four-site spin chain with uniform nearest-neighbour couplings. The dimensionless eigenenergies $\varepsilon_k = E_k/J_{intra}$ are shown as functions of $r = \Delta J/J_{intra}$. Results obtained by numerical diagonalization are shown as blue solid curves, and the corresponding ASN predictions are shown as red dashed curves. At $r = 0$, the eight branches form two fourfold-degenerate manifolds at $\varepsilon = 0$ and $\varepsilon = -2$.

After activation selection and symbolic extraction, every predicted eigenenergy was reduced to a quadratic polynomial containing at most three terms (Eqs. (13)). The resulting expressions have the general form $E_k/J_{intra} = c_{k,0} + c_{k,1}r + c_{k,2}r^2$. Multiplication by the signed J_{intra} restored the physical dimensions while retaining the branch labels defined by the descending dimensionless-energy ordering shown in **Fig. 5**. The resulting analytical approximations are:

$$E_1^{ASN} = -0.016J_{intra} + 0.915\Delta J + 0.128\Delta J^2/J_{intra}, \quad (13a)$$
$$E_2^{ASN} = -0.006J_{intra} + 0.351\Delta J + 0.032\Delta J^2/J_{intra}, \quad (13b)$$
$$E_3^{ASN} = -0.337\Delta J + 0.188\Delta J^2/J_{intra}, \quad (13c)$$
$$E_4^{ASN} = -0.010J_{intra} - 0.772\Delta J + 0.282\Delta J^2/J_{intra}, \quad (13d)$$
$$E_5^{ASN} = -1.991J_{intra} + 0.775\Delta J - 0.284\Delta J^2/J_{intra}, \quad (13e)$$
$$E_6^{ASN} = -2.001J_{intra} + 0.338\Delta J - 0.188\Delta J^2/J_{intra}, \quad (13f)$$
$$E_7^{ASN} = -1.995J_{intra} - 0.349\Delta J - 0.034\Delta J^2/J_{intra}, \quad (13g)$$
$$E_8^{ASN} = -1.979J_{intra} - 0.933\Delta J - 0.118\Delta J^2/J_{intra}. \quad (13h)$$

The coefficients in Eqs. (13a)– (13h) reproduce the qualitative perturbative structure of the four-site Hamiltonian. At $\Delta J = 0$, the constant terms recover the two fourfold-degenerate manifolds at $E_k/J_{intra} = 0$ and $E_k/J_{intra} = -2$, with deviations not exceeding 0.021. To first order in ΔJ, diagonalization of either tridiagonal exchange block gives the linear coefficients $\cos(k\pi/5)$, with $k \in \{1, \dots, 4\}$, corresponding approximately to ±0.809 and ±0.309. The coefficients multiplying ΔJ in Eqs. (13a)–(13h) follow the same sequence of signs and magnitudes but do not coincide exactly with these local perturbative values because the ASN minimizes the global approximation error over the complete sampled interval $0 \leq \Delta J/J_{intra} \leq 3$. The terms proportional to $\Delta J^2/J_{intra}$ describe the curvature generated by double-quantum coupling between the two manifolds.

The exact eigenenergy set is symmetric about $E_k/J_{intra} = -1$, such that each pair of branches E_k and E_{9-k} sums to $-2J_{intra}$ for $k = 1, \dots, 4$. Although this constraint was not imposed during training, Eqs. (13a)–(13h) reproduce it with a maximum deviation of approximately $0.041|J_{intra}|$ over the sampled interval. A symmetry-constrained ASN could enforce the relation exactly by predicting four branches and constructing the remaining four algebraically.

D. Independent verification and quadratic baseline

To provide a controlled comparison with fixed-basis quadratic regression, we repeated the four-site training using an independently written ASN implementation and evaluated both approaches on the same 10,000-parameter sample and 80/20 train-test split. The network architecture and activation set were unchanged from those described in Sec. V C, and the complete training settings are given in Methods. The independent run yielded an RMSE of 7.53×10^{-3}, close to the corresponding value of 7.9×10^{-3} obtained in the primary calculation. The small difference reflects the use of a separate training and evaluation run. The coefficient of determination R^2 ranged from 0.9935 to 0.9998 across the eight branches, and all 11 activation-selection coefficient vectors had a support size of one after the transition to sparsemax.

On the same parameter sample and train-test split, ordinary least-squares regression using the quadratic basis $\{1, r, r^2\}$ gives a slightly lower RMSE of 7.02×10^{-3}. All errors refer to the dimensionless eigenenergies E_k/J_{intra}. Representative fitted expressions are

$$E_1^{LS} = -0.014J_{intra} + 0.908\Delta J + 0.133\Delta J^2/J_{intra}, \quad (14a)$$
$$E_2^{LS} = -0.005J_{intra} + 0.349\Delta J + 0.034\Delta J^2/J_{intra}, \quad (14b)$$

which closely match the corresponding ASN expressions in Eqs. (13a) and (13b). The independent calculation therefore confirms both the numerical accuracy of the four-site ASN and the emergence of one-sparse activation selections. It does not, however, provide a predictive advantage over regression with a fixed quadratic basis. The four-site problem nevertheless provides a nontrivial multi-output test of reproducibility.

VI. Practical considerations and limitations

The ASN architecture should be chosen according to the expected complexity of the target expression. Network width provides independently weighted intermediate variables, whereas depth increases compositional complexity. With the candidate set {0, x, x^2}, the maximum polynomial degree can grow as 2^L across L consecutive ASN layers, while mixed terms arise from applying quadratic activations to linear combinations of several variables.

When a suitable low-order polynomial basis is known in advance, ordinary least-squares regression provides the natural baseline: it is faster, globally optimized, and immediately symbolic. The ASN is intended for problems in which the appropriate analytical basis is uncertain and must be selected from several candidate functions. The choice of this library determines the class of expressions that can be recovered.

Energy branches must be labelled consistently near degeneracies and avoided crossings. To assess the consistency of convergence, we performed five training runs with different random starting values for the network parameters. Known physical relations between eigenenergies can also be incorporated into the model. For example, when the sum of a group of eigenenergies is known from the Hamiltonian, one energy can be calculated from the others. This reduces the number of outputs that must be learned independently and ensures that the predicted energies satisfy the imposed physical constraint.

VII. Conclusions

We introduced the activation-selection network as a differentiable symbolic-regression framework in which each node learns a sparse selection from a predefined library of analytical functions. The resulting expressions can be extracted directly from the trained network. Compositions of the candidate functions {0, x, x^2} generate mixed polynomial terms and progressively higher degrees with increasing network depth.

For symbolic regression of physical quantities, we recommend normalizing the inputs and targets by appropriate physical scales before optimization. Directly combining powers of dimensional inputs can otherwise produce sums of terms with incompatible units, such as Hz and Hz². Restoring the output scale after regression ensures dimensional homogeneity: every term in the extracted physical expression then has the required units.

We demonstrated that ASN can learn analytical approximations to the eigenvalues of parameter-dependent Hamiltonians from numerical data. We considered a 4 × 4 Hamiltonian matrix depending on two independent parameters for the three-site spin chain and an 8 × 8 matrix depending on one parameter for the four-site chain. For the three-site chain, the learned quadratic expressions reproduce the qualitative structure of second-order perturbation theory along the uniform-chain line, while the nonpolynomial splitting in the nonuniform case highlights a limitation of the chosen polynomial function library. For the four-site chain, ASN approximates all eight eigenenergy branches simultaneously.

Fixed-basis quadratic least-squares regression was nevertheless slightly more accurate for both benchmarks when the appropriate polynomial basis was specified in advance. The central contribution of the ASN is therefore not a predictive advantage over regression with a known basis, but the ability to make the selection and composition of candidate analytical functions trainable and to extract compact global approximations directly from numerical data. Stronger evidence for adaptive function selection will require benchmarks with competing nonpolynomial primitives, multiple training seeds, structured extrapolation tests, symmetry-constrained outputs, and direct comparisons with established symbolic-regression methods.

VIII. Methods

A. Numerical dataset generation

For the three-site chain, the dimensionless coupling ratios $r_1 = \Delta J_1/J_{intra}$ and $r_2 = \Delta J_2/J_{intra}$ defined in Eq. (11) were sampled independently and uniformly over $0 \le r_1, r_2 \le 2$. A total of 100,000 parameter pairs were generated using NumPy random seed 42. For each pair, the 4 × 4 Hamiltonian block shown in **Fig. 2(a)** was diagonalized directly. The eigenvalues were divided by the signed J_{intra} and ordered according to $\varepsilon_1 > \varepsilon_2 \ge \varepsilon_3 \ge \varepsilon_4$, as defined in Sec. IV B. Because $J_{intra} < 0$ for the physical systems considered here, this is the reverse of the ordering of the signed dimensional eigenenergies.

For the primary four-site dataset, J_{intra} and ΔJ were sampled independently over their respective intervals. Consequently, the derived dimensionless ratio $r = \Delta J/J_{intra}$ followed a nonuniform distribution spanning approximately $0 \le r \le 3$. Numerical eigenenergies were first calculated on a 200 × 200 grid spanning $-32 \le J_{intra} \le -5$ and $-15 \le \Delta J \le -0.1$ in frequency units. Continuous eigenenergy branches were tracked between neighbouring grid points by maximizing the overlap of the corresponding eigenvectors. Linear interpolation of the resulting branch surfaces was then used to generate 10,000 examples with J_{intra} and ΔJ sampled independently over these intervals. After normalization, the eight outputs $\varepsilon_k = E_k/J_{intra}$ were ordered from highest to lowest.

The independent four-site verification used a distinct dataset, described in Sec. VIII E, for which the 8 × 8 Hamiltonian was diagonalized directly at every sampled value. The primary and independent calculations therefore constitute separate training and evaluation runs rather than repeated evaluations of an identical fitted model.

B. Network training and architecture selection

All ASN models were implemented in Python using PyTorch. Each ASN layer used the candidate activation set $\{0, x, x^2\}$ and included a linear bias. During the first 90% of the training epochs, activation-selection coefficients were calculated using softmax Eq. (1). The $L_{0.5}$-type term in Eq. (3), multiplied by a coefficient of 0.1, was added to the fitting loss to encourage coefficient vectors close to a simplex vertex. During the final 10% of training, softmax was replaced by sparsemax Eq. (4) and the regularization term was removed. Sparsemax permits exact zeros but does not itself guarantee a support size of one.

For the three-site problem, candidate depths and widths were compared using prediction error and symbolic complexity, defined as the total number of arithmetic operations in the extracted expressions. The non-dominated candidates formed a Pareto front, from which the architecture (2, 16, 4) was selected as a compact model. Here and below, an architecture tuple lists the numbers of input, hidden, and output nodes. The selected model was trained for 1000 epochs using the Adam optimizer, a learning rate of 10^{-3}, a batch size of 4096, and mean absolute error as the primary fitting loss. Further details of the Pareto selection and the polynomial structure generated by this architecture are given in Appendix C.

For the primary four-site calculation, the architecture was (1, 10, 8). The model was trained for 5000 epochs using Adam with a learning rate of 10^{-3}, a batch size of 8000, and mean-squared error as the primary fitting loss. For both primary benchmarks, the data were divided randomly into 80% training and 20% validation subsets using PyTorch random seed 42.

C. Symbolic extraction and model evaluation

After training, the activation-selection coefficients, linear weights, and biases were converted directly into symbolic expressions. Consecutive ASN layers were composed algebraically, and the resulting expressions were expanded and simplified using SymPy. Coefficients with absolute values below 10^{-3} were removed, and the remaining coefficients were rounded to three decimal places.

For the three-site chain, coefficients multiplying symmetry-equivalent terms were averaged after extraction because the interchange symmetry $r_1 \leftrightarrow r_2$ was not imposed explicitly during training. Multiplication of the dimensionless expressions by the signed J_{intra} restored physical dimensions and produced Eqs. (12a)–(12d). The same dimensional restoration was applied to the four-site expressions in Eqs. (13a)–(13h). Thus, the published formulas include two explicit post-processing operations: symmetry averaging where applicable and coefficient thresholding and rounding.

Prediction accuracy was quantified using RMSE and the coefficient of determination R^2. The overall RMSE was calculated as the square root of the mean squared residual over all evaluated samples and output branches, whereas R^2 was calculated separately for each branch. Metrics for the original neural runs were evaluated before symbolic thresholding and rounding. Rounded

analytical expressions and independently reproduced neural models were evaluated separately so that training error and symbolic post-processing error were not conflated. The support size of an activation-selection coefficient vector was defined as the number of nonzero coefficients after the transition to sparsemax.

D. Fixed-basis regression baselines

The three-site fixed-basis models used the same 100,000 dimensionless input pairs and the same 80/20 split as the corresponding ASN calculation. Each eigenenergy surface was fitted independently by ordinary least squares. The symmetry-constrained quadratic basis was $\{1, r_1 + r_2,\ r_1^2 + r_2^2,\ r_1 r_2\}$. The physics-informed basis additionally contained the radial feature $\rho = \sqrt{r_1^2 + r_2^2}$. Coefficients were determined without regularization. Fits to the complete dataset were used to compare coefficients with Eqs. (12a)–(12d), whereas held-out errors were calculated on the validation subset.

To examine the local behavior near the degeneracy, the globally fitted three-site models were evaluated without refitting on a 101 × 101 grid over $0 \leq r_1, r_2 \leq 0.2$. The perturbative expressions in Eqs. (10a)–(10d) were evaluated on the same grid.

For the four-site baseline, each of the eight dimensionless eigenenergy branches was fitted independently using the basis $\{1, r, r^2\}$. The least-squares models, the independently trained ASN, and the rounded expressions in Eqs. (13a)–(13h) were evaluated using the same parameter sample and train-test split. Representative least-squares expressions are given in Eqs. (14a) and (14b).

E. Independent verification runs

The reduced three-site verification used 20,000 independently generated samples, architecture (2, 16, 4), 400 training epochs, mean absolute error, a learning rate of 10^{-3}, a batch size of 4096, and the same regularization coefficient and softmax-to-sparsemax schedule as the primary calculation. This reduced run was used only to test numerical reproducibility and the final support sizes; it was not used to obtain Eqs. (12a)–(12d) or the primary three-site error values.

For the independent four-site verification, 10,000 values of r were sampled uniformly over $0 \leq r \leq 3$ using NumPy random seed 42. The eight dimensionless target eigenenergies were obtained by direct diagonalization of the 8 × 8 Hamiltonian at each sampled value, rather than by interpolation of a precomputed surface. The data were divided using an 80/20 split generated with PyTorch random seed 42. The ASN architecture and optimization settings were otherwise those specified for the four-site model in Sec. VIII B. The quadratic baseline was trained and evaluated on the same split. This controlled protocol produced the values reported in Sec. V D.

Appendices

Appendix A: Mathematical details of the activation-selection network and multiplication example

Figure 1(b) illustrates a two-layer ASN that multiplies two arbitrary inputs x_1 and x_2 using the candidate set $\{0, x, x^2\}$. The zero activation suppresses unused paths. In the first layer, both input nodes select the identity activation. A convenient set of logits and the resulting sparse coefficient vectors is:

$$\boldsymbol{\alpha}^{(1)} = \begin{bmatrix} -50 & 50 & -50 \\ -50 & 50 & -50 \end{bmatrix}. \tag{A1}$$

$$\mathbf{a}^{(1)} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \qquad v_i^{(1)} = x_i. \tag{A2}$$

The large positive and negative logits are illustrative finite values whose sparsemax projections select the indicated functions. The corresponding weights and biases are

$$\mathbf{w}^{(1)} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \qquad \mathbf{b}^{(1)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \tag{A3}$$

and therefore

$$y_1 = x_1 + x_2, \qquad y_2 = x_1 - x_2. \tag{A4}$$

In the second layer, both intermediate nodes select the quadratic activation,

$$\boldsymbol{\alpha}^{(2)} = \begin{bmatrix} -50 & -50 & 50 \\ -50 & -50 & 50 \end{bmatrix}. \tag{A5}$$

$$\mathbf{a}^{(2)} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \qquad v_i^{(2)} = y_i^2. \tag{A6}$$

with the final linear transformation

$$\mathbf{w}^{(2)} = [1/4 \quad -1/4], \qquad b^{(2)} = 0. \tag{A7}$$

which gives

$$z_1 = 1/4\, y_1^2 - 1/4\, y_2^2 = x_1 x_2. \tag{A8}$$

Thus, the ASN implements multiplication exactly. Repeated composition of the same candidate functions can generate higher-degree polynomials, although the topology realizing a given expression is not unique.

Appendix B: Effective-spin basis and construction of the benchmark Hamiltonian matrices

The Hamiltonian acts within the reduced Hilbert space spanned by direct products of the singlet and central-triplet states,

$$B^{ST} = \{|T_0\rangle^{(1)}, |S_0\rangle^{(1)}\} \otimes \{|T_0\rangle^{(2)}, |S_0\rangle^{(2)}\} \otimes \ldots \otimes \{|T_0\rangle^{(N)}, |S_0\rangle^{(N)}\} \quad \text{(B1)}$$

Within this reduced representation, the intrapair interaction appears as an effective Zeeman term proportional to $\hat{I}_{i,z}$, which lifts the degeneracy between the $|T_0\rangle^{(i)}$ and $|S_0\rangle^{(i)}$ states of the i-th CH_2 group. The identification $|\uparrow\rangle \equiv |T_0\rangle$ and $|\downarrow\rangle \equiv |S_0\rangle$ is used throughout the manuscript. The effective couplings ΔJ_i describe nearest-neighbour interactions between adjacent effective spins. In the original aliphatic chain, these couplings are given by the difference between the two conformationally averaged vicinal three-bond *J*-couplings (see **Fig. B1**):

$$\Delta J_i \equiv \langle J_{i,i+3} \rangle - \langle J_{i,i+2} \rangle \quad \text{(B2)}$$

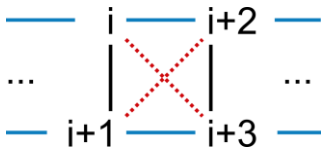


Fig. B1. Proton-numbering convention for two neighboring methylene groups. Protons i and $i+1$ form the first pair, while $i+2$ and $i+3$ form the second. Here, i labels individual protons rather than effective-spin sites.

The first term in Eq. (8) dominates under realistic conditions because the intrapair coupling $|J_{intra}|$ is substantially larger than all effective couplings $|\Delta J_i|$. Its eigenstates are direct-product states in the effective spin basis, such as $|\uparrow\uparrow\uparrow\downarrow\rangle$. The corresponding eigenenergies are obtained by assigning an energy contribution of $J_{intra}/4$ to each spin-up state and $-3J_{intra}/4$ to each spin-down state and summing these contributions over all sites:

$$E^{(0)} = \left(\frac{n_\uparrow}{4} - \frac{3n_\downarrow}{4}\right) J_{\text{intra}}, \qquad n_\uparrow + n_\downarrow = N. \quad \text{(B3)}$$

The exchange and double-quantum terms in Eq. (8) both generate matrix elements of $\Delta J_i/2$. Exchange terms connect product states differing by $|\cdots\uparrow\downarrow\cdots\rangle \leftrightarrow |\cdots\downarrow\uparrow\cdots\rangle$, thereby lifting their degeneracy and substantially modifying the eigenstates. This term is equivalent to the nearest-neighbour XY spin-chain Hamiltonian, whose eigenstates and eigenfrequencies are known analytically [9].

The double-quantum terms connect states differing by two spin flips: $|\cdots\uparrow\uparrow\cdots\rangle \leftrightarrow |\cdots\downarrow\downarrow\cdots\rangle$. Although the exchange and double-quantum matrix elements have the same magnitude, the exchange term acts within degenerate manifolds and contributes at first order. By contrast, the double-quantum term couples manifolds separated by an energy of order $2|J_{intra}|$ and therefore contributes to the eigenenergies only at second order when $|\Delta J_i|/2 \ll 2|J_{intra}|$. Its effect can consequently be treated perturbatively [8,9,12].

The exchange interaction preserves the number of |↓⟩ sites, whereas the double-quantum interaction changes it by ±2. Both interactions therefore preserve its parity, and the Hamiltonian separates into even- and odd-parity blocks, each of dimension 2^{N-1}. For the three-site benchmark, **Fig. 2(a)** shows the four-dimensional even-parity block, comprising one state with no |↓⟩ sites and three states with two |↓⟩ sites. For the four-site benchmark, **Fig. 2(b)** shows the eight-dimensional odd-parity block, comprising four states with one |↓⟩ site and four states with three |↓⟩ sites.

Appendix C: Architecture selection and construction of the extracted polynomial expressions

Candidate three-site architectures with different depths and widths were trained using the same activation library. Each candidate was characterized by its prediction error and by the number of arithmetic operations in the extracted symbolic expressions after coefficient thresholding. A candidate belonged to the Pareto front when no other model had both lower error and lower symbolic complexity. The architecture (2, 16, 4) was retained because it provided a compact point on this front while preserving the accuracy required for the subsequent symbolic analysis.

Although the selected model contains one hidden layer, it comprises two consecutive activation-selection stages. Consider a hidden variable formed after identity selection in the first stage,

$$h_s = w_{s1} r_1 + w_{s2} r_2 + b_s. \tag{C1}$$

If this variable selects the quadratic activation in the following stage, its contribution becomes

$$h_s^2 = w_{s1}^2 r_1^2 + 2 w_{s1} w_{s2} r_1 r_2 + w_{s2}^2 r_2^2 + 2 b_s w_{s1} r_1 + 2 b_s w_{s2} r_2 + b_s^2. \tag{C2}$$

A single quadratic activation therefore generates r_1^2, r_2^2, and the mixed term $r_1 r_2$, together with linear and constant contributions from the bias. Other hidden variables selecting the identity function provide additional linear terms, while zero activations suppress unnecessary paths. Linear combinations of these hidden-node outputs produce the complete quadratic form recovered in Eqs. (12a)–(12d). If a first-stage quadratic output is squared again, the same architecture can generate quartic terms, although such terms were not retained in the extracted benchmark expressions.

Appendix D: Two-dimensional degenerate perturbation theory for the nonuniform three-site chain

Within the even-parity block, let P project onto the three-dimensional manifold spanned by {|↑↓↓⟩, |↓↑↓⟩, |↓↓↑⟩}, with unperturbed energy $A = -5J_{intra}/4$, and let Q project onto the remaining state |↑↑↑⟩, with energy $B = 3J_{intra}/4$. In this ordered basis, the first-order perturbation is:

$$V_P = \begin{bmatrix} 0 & d_1 & 0 \\ d_1 & 0 & d_2 \\ 0 & d_2 & 0 \end{bmatrix}, \qquad d_i = \frac{\Delta J_i}{2}, \qquad R = \sqrt{d_1^2 + d_2^2}. \tag{D1}$$

The first-order eigenvalues are $+R$, 0, and $-R$, with normalized eigenvectors

$$|+\rangle = \frac{1}{\sqrt{2}R}\begin{bmatrix} d_1 \\ R \\ d_2 \end{bmatrix}, |0\rangle = \frac{1}{R}\begin{bmatrix} d_2 \\ 0 \\ -d_1 \end{bmatrix}, \quad |-\rangle = \frac{1}{\sqrt{2}R}\begin{bmatrix} d_1 \\ -R \\ d_2 \end{bmatrix}. \tag{D2}$$

The Q-manifold state couples to P through the vector and squared matrix elements

$$\mathbf{v}_{QP} = [d_2 \quad 0 \quad d_1], \quad |\langle Q|V|\pm\rangle|^2 = \frac{2d_1^2 d_2^2}{R^2}, \quad |\langle Q|V|0\rangle|^2 = \frac{\left(d_2^2 - d_1^2\right)^2}{R^2}. \tag{D3}$$

Using $A - B = -2J_{intra}$ gives the second-order corrections within P,

$$\delta E_{\pm}^{(2)} = -\frac{d_1^2 d_2^2}{J_{intra} R^2}, \qquad \delta E_0^{(2)} = -\frac{\left(d_2^2 - d_1^2\right)^2}{2J_{intra} R^2}. \tag{D4}$$

whereas the Q-manifold state receives

$$\delta E_Q^{(2)} = \frac{R^2}{2J_{\mathrm{intra}}}. \tag{D5}$$

Substituting $d_i = \Delta J_i/2$ and $S = \Delta J_1^2 + \Delta J_2^2$ yields Eqs. (10a)– (10d). At $S = 0$, the rational contributions are assigned their continuous zero-coupling limits. Along the uniform branch, $\Delta J_1 = \Delta J_2 = \Delta J$, with $\Delta J/J_{\mathrm{intra}} \geq 0$ these expressions reduce to Eqs. (9a)–(9d) through second order. When the sign of ΔJ is reversed, the two split branches exchange their order.

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Data Availability Statement

The code and datasets supporting the findings of this study are available in the NMRKan repository: A. Yu. Kharin, *NMRKan: NMR symbolic regression repository*, GitHub (2026), https://github.com/Alexankharin/NMRKan

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